HYPERSONICS / CFD

Shock-shock interactions at Mach 3

Shock-polar classification and CFD of regular and Mach interactions for opposed ramps at ISA 20 km freestream conditions.

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Python code: shockPolar.py

Separated from the original Hypersonics and Advanced Propulsion report. Equations, result figures and technical text retain their original numbering. Shared setup material is included where needed. No simulation results have been changed.

Table 1: Freestream conditions (ISA 20 km, $\gamma = 1.4$, $C_p = 1004.5\,\mathrm{J/(kg\,K)}$).

Quantity	Symbol	Value	Unit
Static pressure	p_∞	5529	Pa
Density	ρ_∞	0.088	$\mathrm{kg/m^3}$
Temperature	T_∞	216.65	K
Speed of sound	a_∞	≈ 295	m/s
Dynamic viscosity	μ	$\approx 1.42 \times 10^{-5}$	Pa·s

2.2. Shock Polar

A shock polar traces all oblique shock solutions in the $(\theta,\, p/p_\infty)$ plane:

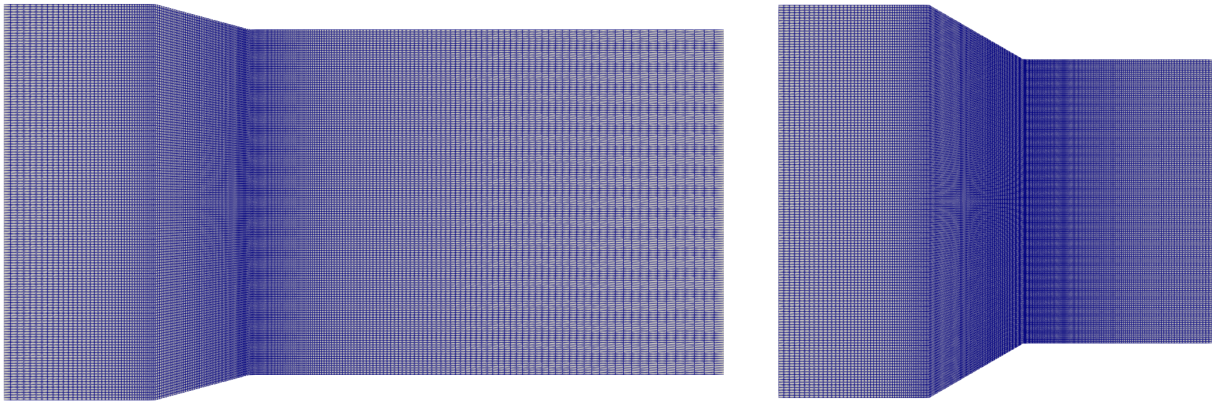
$$\frac{p_2}{p_1} = 1 + \frac{2\gamma}{\gamma+1}(M_1^2 \sin^2\beta - 1), \qquad \tan\theta = \frac{2\cot\beta(M_1^2\sin^2\beta - 1)}{M_1^2(\gamma + \cos 2\beta) + 2}. \tag{3}$$

A **Type I** (regular) interaction occurs when the two downstream polars intersect on the weak-shock branch. A **Type II** (Mach interaction) occurs when no intersection exists and a Mach stem forms.

Both Mach 3 ramp cases use rhoCentralFoam with Kurganov-Tadmor flux, van Leer reconstruction, a maximum Courant number of 0.4, and slip walls.

The Type I case has opposed 15-degree ramps and approximately 33,000 cells; the Type II case has opposed 30-degree ramps and approximately 40,500 cells.

The Kantrowitz area ratios are 1.147 and 1.38, respectively, both below the 1.39 starting limit.

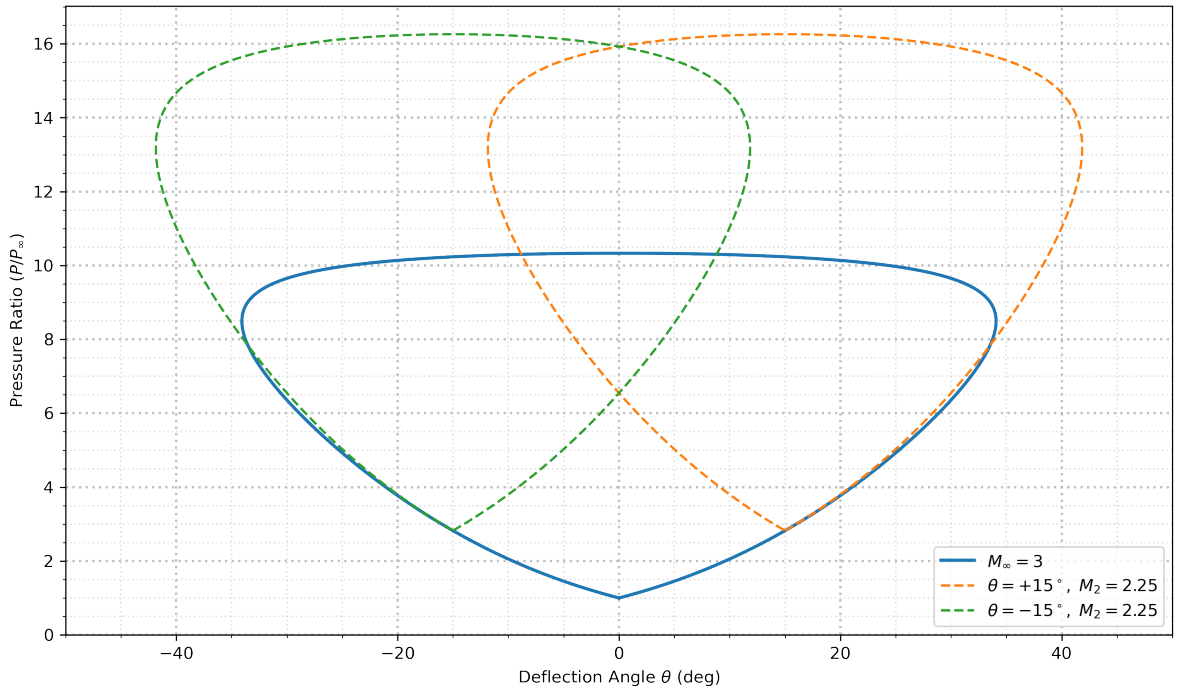


(a) type I interaction mesh. (b) type II interaction mesh.

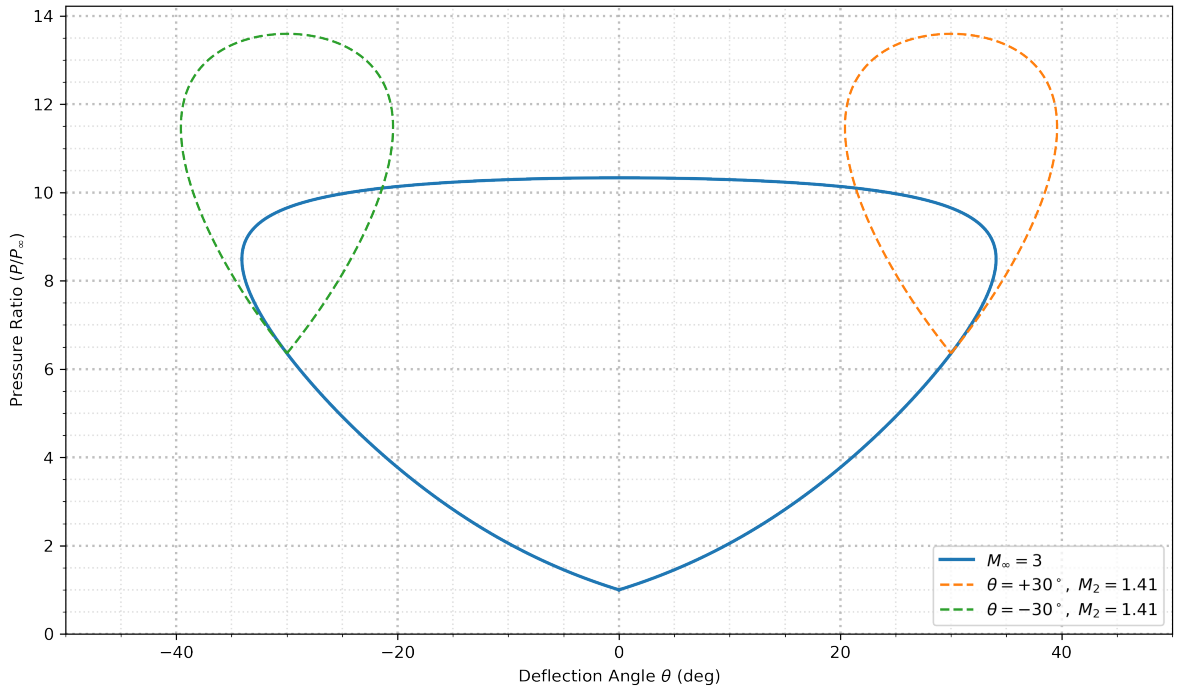
Ramp meshes for the Type I and Type II interaction cases.

5. Shock-Shock Interaction Results

5.1. Polar Classification



(a) Type I ($\theta = \pm 15°$), regular interaction.



(b) Type II ($\theta = \pm 30°$), Mach interaction.

Figure 9: Shock polars at $M_\infty = 3$.

5.2. Type I CFD Results

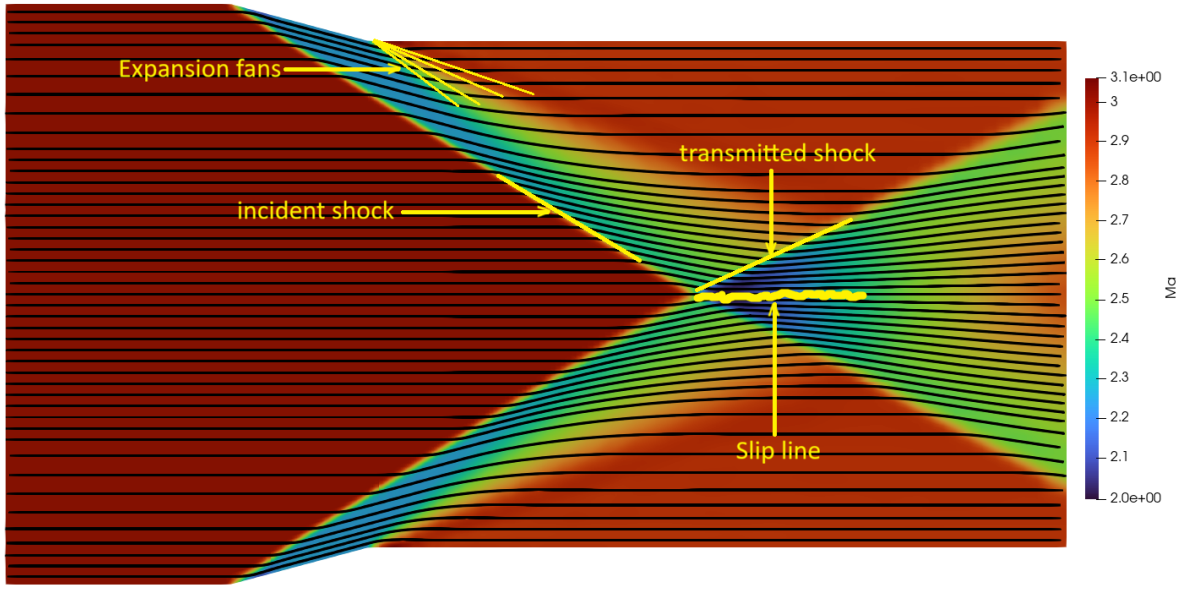
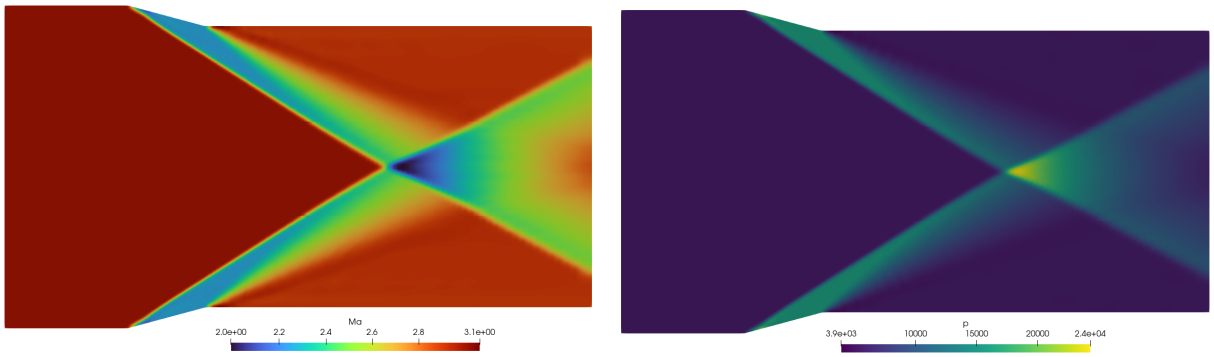


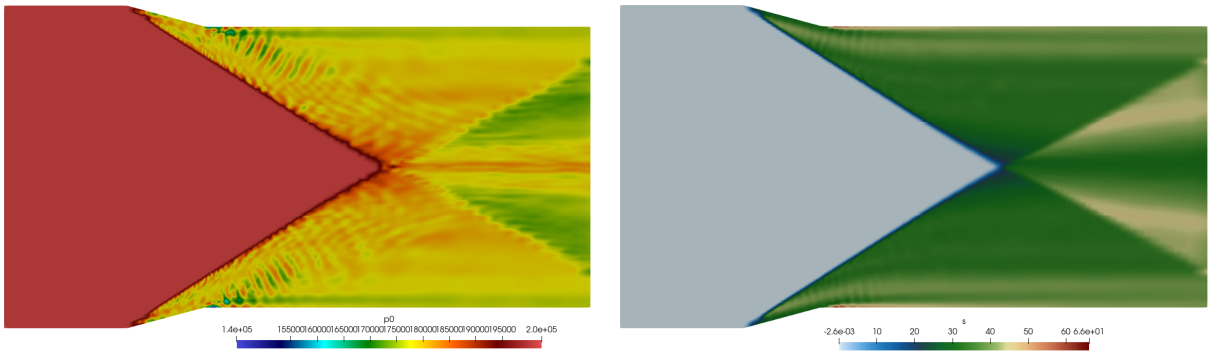
Figure 10: streamlines and flow features in regular interaction.



(a) Mach number.

(b) Static pressure.

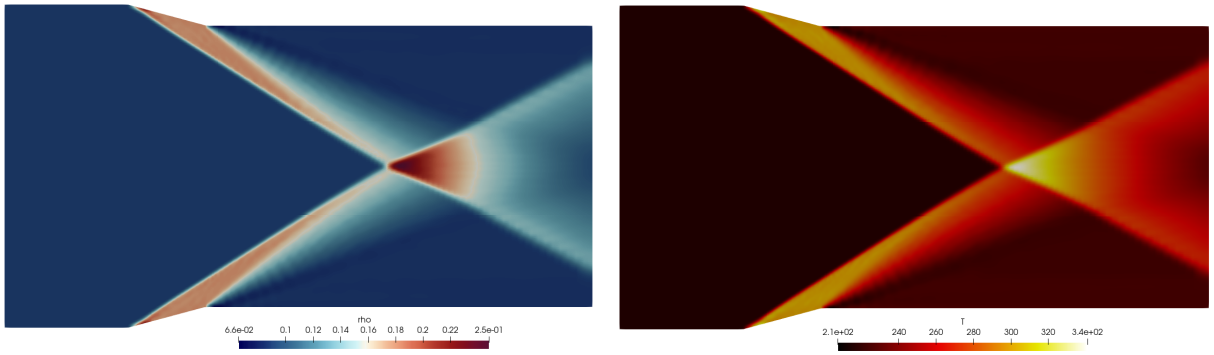
Figure 11: Type I flow field.



(a) Total pressure.

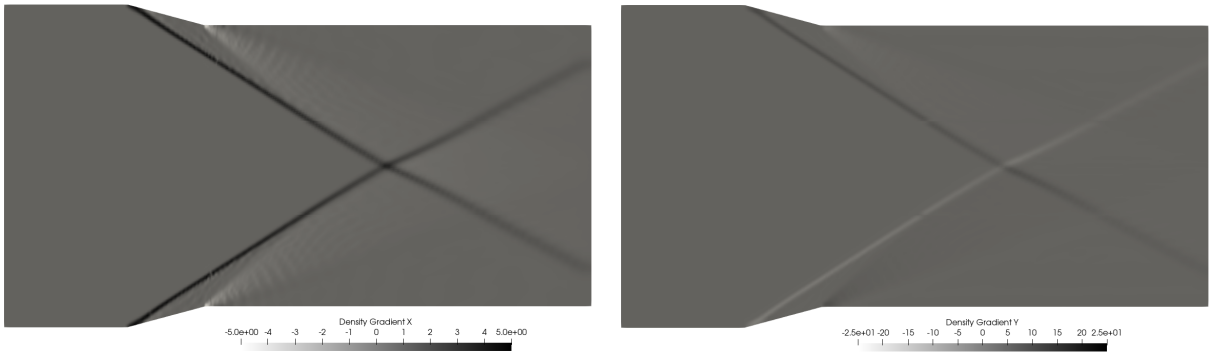
(b) Entropy.

Figure 12: Type I total pressure and entropy.



(a) Density. (b) Temperature.

Figure 13: Type I density and temperature.



(a) $\partial\rho/\partial x$. (b) $\partial\rho/\partial y$.

Figure 14: Numerical Schlieren for the Type I case.

The two incident shocks cross symmetrically and produce reflected shocks without a Mach stem. A narrow slip line is visible at the crossing point in the entropy contour (Figure 12b). The total pressure and entropy contours both show this slipline passing through the centreline.

5.3. Type II CFD Results

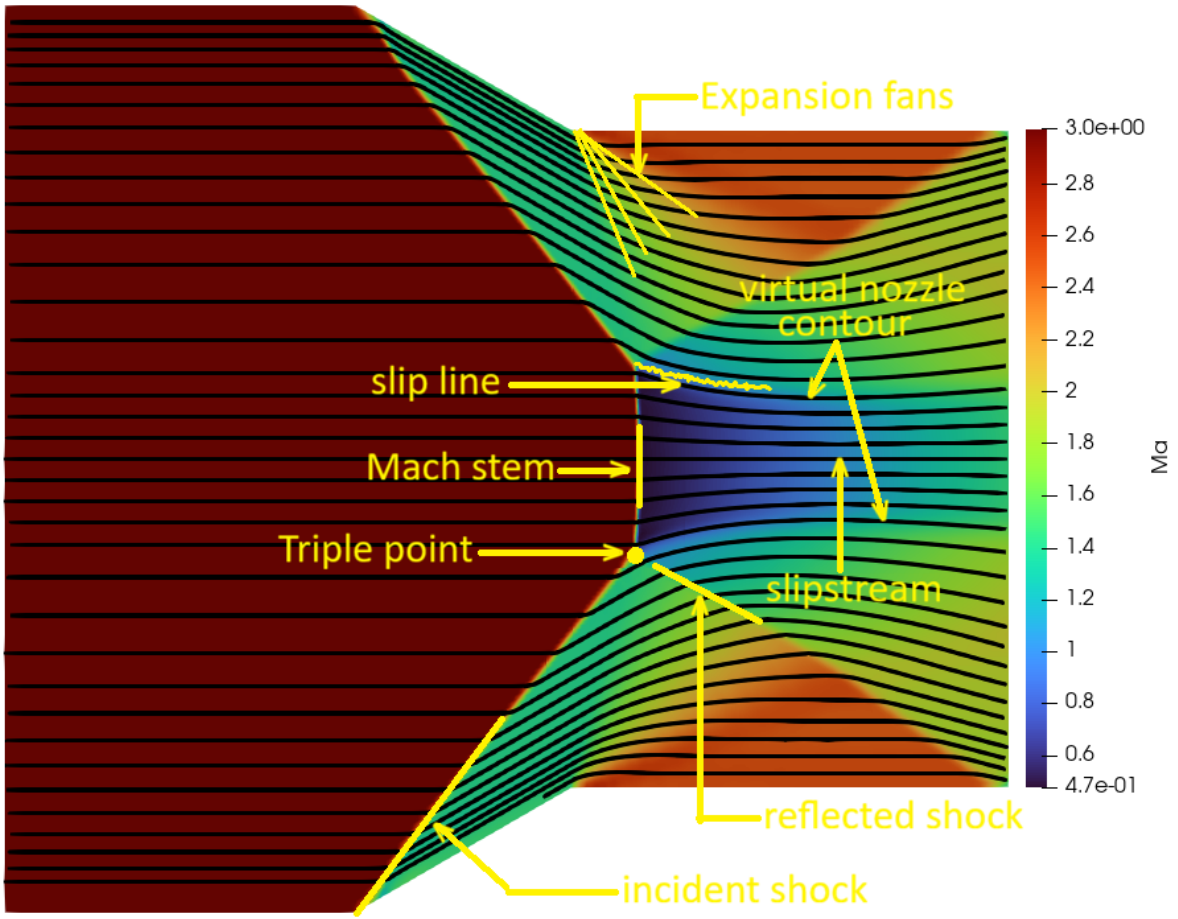
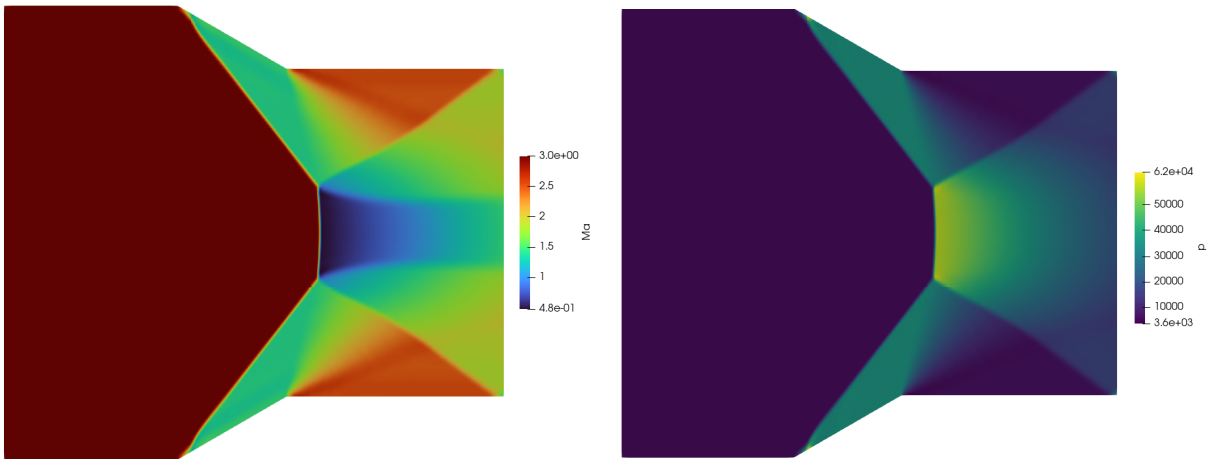


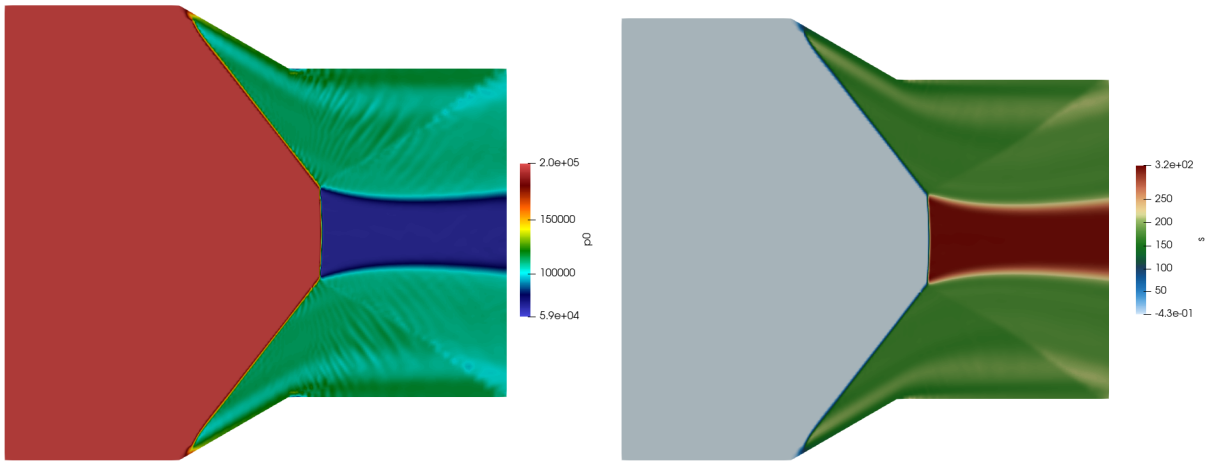
Figure 15: streamlines and flow features in Mach interaction.



(a) Mach number.

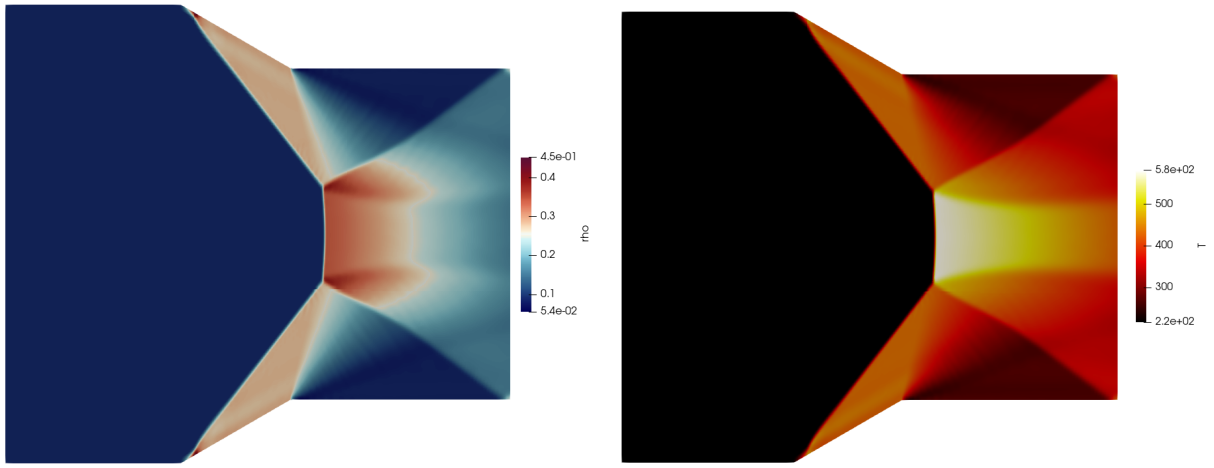
(b) Static pressure.

Figure 16: Type II flow field.



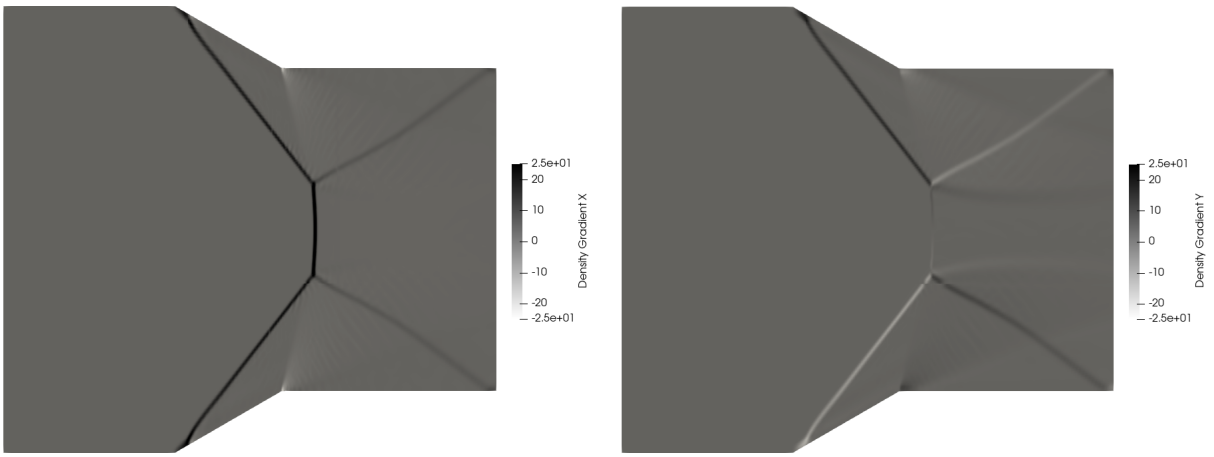
(a) Total pressure. (b) Entropy.

Figure 17: Type II total pressure and entropy.



(a) Density. (b) Temperature.

Figure 18: Type II density and temperature. The density jump across the Mach stem is substantially larger than across the oblique shocks.



(a) $\partial\rho/\partial x$. (b) $\partial\rho/\partial y$.

Figure 19: Numerical Schlieren for the Type II case.

A near-normal Mach stem spans the channel centreline, producing a subsonic pocket that is clearly visible in the Mach contour. Reflected shocks leave the stem and propagate downstream. A slipstream extends downstream from each triple point. The Mach stem generates a substantially larger entropy rise and total pressure loss than the oblique shocks. Expansion fans at the triple points then interact with the downstream flow, partially recovering the Mach number.

6. Discussion

Shock-shock interactions. CFD confirms the polar predictions at both deflection angles. The Type I case shows a clean regular reflection with negligible total pressure loss. The Type II case exhibits all features of a Mach interaction: a near-normal Mach stem, two triple points, a slipstream separating subsonic and supersonic regions, and expansion fans at the triple points. The Kantrowitz limit check confirms that the duct can start without a normal shock. Without satisfying this criterion, a standing normal shock at the ramp entrance would choke the duct.

Listing 2: Shock polar generation (`shockPolar.py`).

```
import numpy as np
import matplotlib.pyplot as plt

GAMMA = 1.4
M_inf = 3
deflections = [15, -15]

def shock_polar(M, theta_offset=0.0, n_points=1000):
    mu = np.degrees(np.arcsin(1 / M))
    beta = np.linspace(mu, 90.0, n_points)
    b_rad = np.radians(beta)
    p_ratio = 1 + (2 * GAMMA / (GAMMA + 1)) * ((M * np.sin(b_rad))**2 - 1)
```

```
    theta_rel = np.degrees(
        np.arctan(
            (2 / np.tan(b_rad)) *
            ((M**2 * np.sin(b_rad)**2 - 1) /
             (M**2 * (GAMMA + np.cos(2 * b_rad)) + 2))
        )
    )
    theta_abs = np.concatenate([
        theta_offset + theta_rel,
        theta_offset - theta_rel[::-1]
    ])
    p_full = np.concatenate([p_ratio, p_ratio[::-1]])
    return theta_abs, p_full

def weak_shock_angle(M, theta_deg):
    mu = np.degrees(np.arcsin(1 / M))
    betas = np.linspace(mu, 90.0, 10000)
    for b in betas:
        b_rad = np.radians(b)
        t = np.degrees(
            np.arctan(
                (2 / np.tan(b_rad)) *
                ((M**2 * np.sin(b_rad)**2 - 1) /
                 (M**2 * (GAMMA + np.cos(2 * b_rad)) + 2))
            )
        )
        if t >= theta_deg:
            return b
    return None

def downstream_mach(M, beta, theta):
    b_rad = np.radians(beta)
    Mn1 = M * np.sin(b_rad)
    Mn2 = np.sqrt((1 + (GAMMA - 1) / 2 * Mn1**2) /
                  (GAMMA * Mn1**2 - (GAMMA - 1) / 2))
    return Mn2 / np.sin(np.radians(beta - theta))

fig, ax = plt.subplots(figsize=(10, 6))
theta0, p0 = shock_polar(M_inf, 0.0)
plt.plot(theta0, p0, '-', color='C0', linewidth=2, label=f'$M_\infty={M_inf}$')

colors = ['C1', 'C2']
for i in range(len(deflections)):
    beta = weak_shock_angle(M_inf, abs(deflections[i]))
    if beta is None:
        continue
    M2 = downstream_mach(M_inf, beta, abs(deflections[i]))
    if M2 >= 1.0:
        b_rad = np.radians(beta)
        p_ratio_at_shock = 1 + (2 * GAMMA / (GAMMA + 1)) * ((M_inf * np.sin(
            b_rad))**2 - 1)
        theta2, p2 = shock_polar(M2, theta_offset=deflections[i])
```

Appendix: shock-polar code (continued)

```
        plt.plot(theta2, p_ratio_at_shock * p2, '--', linewidth=1.5, color=
            colors[i], label=rf'$\theta={deflections[i]:+d}^\circ,\;M_2={M2:.2f}
            $')

ax.set_xlabel(r'Deflection Angle $\theta$ (deg)')
ax.set_ylabel(r'Pressure Ratio ($P / P_\infty$)')
ax.set_xlim(-50, 50)
ax.set_ylim(bottom=0)
ax.legend(loc='lower right')
ax.minorticks_on()
ax.grid(True, which='major', linestyle=':', linewidth=1.5, alpha=0.8)
ax.grid(True, which='minor', linestyle=':', linewidth=1.0, alpha=0.4)
plt.tight_layout()
plt.savefig("shockPolar_typeI.png", dpi=300, bbox_inches='tight', transparent=
    True)
plt.show()
```